Supplemental Material: Phase diagram of networks grown with triadic closure rules

Alexei Vazquez[1,*]

[1] *Nodes & Links Ltd, Salisbury House, Station Road, Cambridge, CB1 2LA, UK*

(Dated: October 22, 2026)

I. PER-TRIAD RATES

Let every node spawn a newcomer at rate α_+, which links to it, and every open triad close at rate β_+ ($\beta_- = \alpha_- = 0$). The number of nodes grows as $n \propto e^{\alpha_+ t}$, so node ages τ, in units of $1/\alpha_+$, are exponentially distributed, with density $e^{-\tau}$. In the uncorrelated limit each neighbour of node i has on average $m = \langle k(k-1)\rangle/\langle k\rangle$ further neighbours, so i is an endpoint of about $k_i m$ open triads (closed ones are negligible here), and

$$\frac{dk}{d\tau} = 1 + xk, \qquad x = \frac{\beta_+ m}{\alpha_+}, \tag{1}$$

and, with $k(0) = 1$, $k(\tau) = (1+1/x)e^{x\tau} - 1/x$. Averaging over ages,

$$\langle k\rangle = \frac{2}{1-x}, \qquad \langle k^2\rangle = \frac{(1+1/x)^2}{1-2x} - \frac{2(1+1/x)}{x(1-x)} + \frac{1}{x^2}, \tag{2}$$

and $P(k) \sim k^{-1-1/x}$. The second moment is finite only for $x < 1/2$. Self-consistency requires $m(x) = (\langle k^2\rangle - \langle k\rangle)/\langle k\rangle$, which fixes $\beta_+/\alpha_+ = x/m(x)$. This function vanishes at $x = 0$ and at $x = 1/2$, where m diverges, and has a maximum $\beta_+/\alpha_+ = 0.067$ at $x = 0.225$, where $\langle k\rangle = 2.58$. Above it there is no stationary density.

We simulated rule (w) with these rates, the closure probability per event being $\beta_+ W/(\alpha_+ n + \beta_+ W)$. For $\beta_+/\alpha_+ \geq 0.06$ the mean degree runs away within 10^3–5×10^4 nodes. At 0.02 and 0.04 it stays near 2.09 and 2.23 for a while. The plateau at 0.04 corresponds to an effective $u = 1 - 2/\langle k\rangle \approx 0.10$; the run condenses near $n = 10^6$, as in Sec. VI of the main text, after which $W \sim n^{3/2}$, closures outpace arrivals, and $\langle k\rangle$ reaches 2.67 at 1.6×10^6 and keeps rising.

II. DATA NEAR THE BOUNDARY AND IN THE TREE PHASE

The tables and figure below give the data near the condensation boundary and in the tree phase of rule (w) referred to in the main text. All runs use the code in the repository cited there; the data files are `data/rev_points.csv`, `data/rev_brackets.csv` (`brackets.py`), and `data/rev_seeds.csv`, `data/rev_traj.csv` and `data/rev_tree.csv` (`rev.py`); Fig. S1 is drawn by `mkfigs.py`.

The cumulant in Table S2, $U = 1 - \langle P_\triangle^4\rangle/(3\langle P_\triangle^2\rangle^2)$, is 2/3 for a narrow distribution. It stays at 0.66 in the condensed phase and dips at 0.76 and 0.78, where a few seeds percolate and the distribution of $P_\triangle$ is broad and skewed. The scaled variance $n\,\mathrm{var}(P_\triangle)$ peaks at 0.76 at every size and grows as $n^{0.55}$ between 10^4 and 10^5 and as $n^{0.94}$ between 10^5 and 10^6.

The brackets of Table S1 were also tested against the coordinates in which deletion might act through density or through the rates per surviving node. Neither collapses them. At $\langle k\rangle \approx 3$ the boundary lies at 0.73 ($\Delta = 1/2$, $\alpha_- = 0$) and at 0.875 ($\Delta = 1$, $\alpha_-/\alpha_+ = 0.3$), while at $\alpha_-/\alpha_+ = 0.5$ it lies at 0.89 for both $\Delta = 1/2$ ($\langle k\rangle = 2$) and $\Delta = 1$ ($\langle k\rangle = 8/3$). At $\Delta = 1$ the opening rate per net arrival at the boundary, $\beta_-/(\alpha_+ - \alpha_-)$, grows from 3.3 to 62, and the closure rate $\beta_+/(\alpha_+ - \alpha_-)$ from 4.3 to 67, between $\alpha_-/\alpha_+ = 0$ and 0.8. A fit of $1 - (\beta_-/\beta_+)_c$ linear in $1 - \alpha_-/\alpha_+$ with a free constant gives the constant 0.007.

* alexei@nodeslinks.com

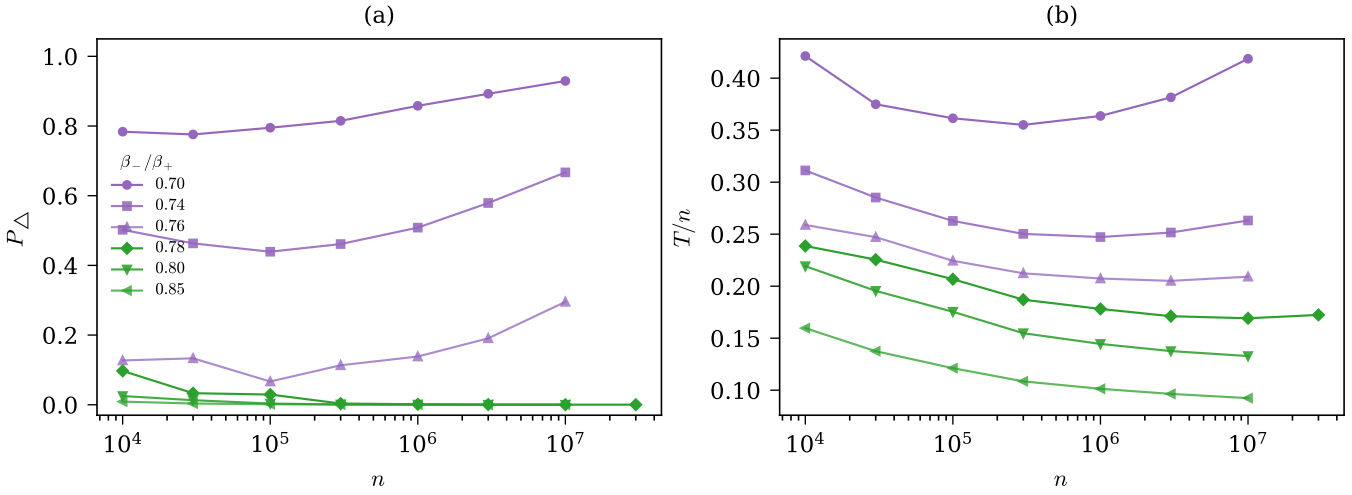


FIG. S1. Single trajectories to $n = 10^7$ (3×10^7 at 0.78) near the boundary, rule (w) at $\Delta = 1$, $\alpha_- = 0$, for the turnovers β_-/β_+ shown; each point is the mean of two seeds, and smaller n are earlier stages of the same runs. Purple: turnovers that condense; green: turnovers that do not up to 10^7. (a) Order parameter $P_\triangle$. (b) Triangles per node T/n.

TABLE S1. Brackets of the condensation boundary of rule (w). For each line of net closure per arrival Δ and deletion α_-/α_+: the largest condensed and the smallest dispersed turnover β_-/β_+, with the seed mean of $P_\triangle$ at $n = 10^5$ and 10^6 and, in brackets, the number of seeds. At $\Delta = 1$, $\alpha_-/\alpha_+ = 0.9$ every point up to 0.95 is condensed.

Δ	α_-/α_+	β_-/β_+	$P_\triangle(10^5)$	$P_\triangle(10^6)$	β_-/β_+	$P_\triangle(10^5)$, $P_\triangle(10^6)$
1/2	0	0.72	0.325 (4)	0.414 (2)	0.74	0.151 (4), 0.091 (2)
1/2	0.5	0.88	0.591 (4)	0.706 (2)	0.9	0.064 (6), 0.001 (3)
1	0	0.76	0.100 (4)	0.138 (2)	0.78	0.021 (4), 0.001 (2)
1	0.1	0.75	0.604 (2)	0.706 (1)	0.8	0.005 (2), 0.001 (1)
1	0.2	0.8	0.161 (2)	0.365 (1)	0.85	0.003 (2), 0.000 (1)
1	0.3	0.85	0.008 (2)	0.039 (1)	0.9	0.002 (2), 0.000 (1)
1	0.4	0.85	0.586 (2)	0.660 (1)	0.9	0.002 (2), 0.000 (1)
1	0.5	0.88	0.206 (4)	0.442 (2)	0.9	0.008 (6), 0.001 (3)
1	0.6	0.9	0.695 (2)	0.699 (1)	0.95	0.002 (2), 0.000 (1)
1	0.7	0.9	0.931 (2)	0.979 (1)	0.95	0.002 (2), 0.000 (1)
1	0.8	0.9	0.990 (2)	0.997 (1)	0.95	0.007 (2), 0.001 (1)

TABLE S2. Statistics of $P_\triangle$ over many seeds near the boundary, $\Delta = 1$, $\alpha_- = 0$: mean and standard error, scaled variance $n\,\mathrm{var}(P_\triangle)$ and cumulant $U = 1 - \langle P_\triangle^4\rangle / (3\langle P_\triangle^2\rangle^2)$, with 64, 32 and 16 seeds at $n = 10^4$, 10^5 and 10^6.

β_-/β_+	n	$P_\triangle$	$n\,\mathrm{var}(P_\triangle)$	U
0.70	10^4	0.7305 ± 0.0063	25.4	0.66
0.70	10^5	0.7773 ± 0.0036	40.7	0.67
0.70	10^6	0.8557 ± 0.0028	114.7	0.67
0.72	10^4	0.6404 ± 0.0081	41.1	0.65
0.72	10^5	0.6584 ± 0.0076	177.9	0.66
0.72	10^6	0.7350 ± 0.0063	594.1	0.67
0.74	10^4	0.4991 ± 0.0119	89.7	0.63
0.74	10^5	0.4559 ± 0.0093	265.5	0.65
0.74	10^6	0.5121 ± 0.0082	1014.7	0.66
0.76	10^4	0.2756 ± 0.0166	173.2	0.45
0.76	10^5	0.1402 ± 0.0140	606.8	0.37
0.76	10^6	0.1369 ± 0.0187	5246.6	0.41
0.78	10^4	0.0895 ± 0.0085	45.3	−0.13
0.78	10^5	0.0095 ± 0.0011	4.0	−0.05
0.78	10^6	0.0013 ± 0.0002	0.4	0.30
0.80	10^4	0.0325 ± 0.0024	3.5	0.13
0.80	10^5	0.0035 ± 0.0002	0.2	0.44
0.80	10^6	0.0004 ± 0.0000	0.0	0.58

TABLE S3. Single trajectories at $\Delta = 1$, $\alpha_- = 0$, mean of two seeds: $P_\triangle$, triangles per node T/n, largest degree $k_{\max}$, and the share of all links that join two of the 1% highest-degree nodes.

β_-/β_+	n	$P_\triangle$	T/n	$k_{\max}$	top-1% links
0.70	10^5	0.7951	0.361	107	0.061
0.70	10^6	0.8580	0.364	208	0.217
0.70	10^7	0.9293	0.419	462	0.444
0.74	10^5	0.4392	0.263	92	0.045
0.74	10^6	0.5084	0.247	174	0.157
0.74	10^7	0.6669	0.263	374	0.396
0.76	10^5	0.0667	0.224	91	0.038
0.76	10^6	0.1384	0.207	163	0.129
0.76	10^7	0.2954	0.209	326	0.358
0.78	10^5	0.0294	0.207	82	0.034
0.78	10^6	0.0014	0.178	144	0.107
0.78	10^7	0.0003	0.169	279	0.313
0.78	3×10^7	0.0002	0.172	408	0.419
0.80	10^5	0.0034	0.175	74	0.027
0.80	10^6	0.0004	0.144	130	0.076
0.80	10^7	0.0001	0.133	227	0.231
0.85	10^5	0.0015	0.121	49	0.013
0.85	10^6	0.0002	0.101	71	0.025
0.85	10^7	0.0000	0.092	102	0.049

TABLE S4. Tree phase ($\beta_- = 2\beta_+$) and tree transition ($\beta_- = \beta_+$) at $\beta_+ = \alpha_+$: range of the largest degree $k_{\max}$ and mean number of triangles T over four seeds.

n	$k_{\max}$, $\beta_- = 2\beta_+$	T	$k_{\max}$, $\beta_- = \beta_+$	T
10^4	9	0.2	$8--11$	74
10^5	$10--13$	1.8	$11--12$	174
10^6	$11--12$	1.2	$11--13$	796
10^7	$12--14$	1.2	$12--13$	2060